

FPGA Synthesis and Prototyping Lab:

35

Binary Adders/Subtractors

CODE:

```
adder.v
/home/mooc-03/Desktop/ADDER/ADDER.srscs/sources_1/new/adder.v

22
23 module adder(
24     input a , b , c ,
25     output sum , carry
26
27 );
28     assign sum = a ^ b ^ c ;
29     assign carry = (a&b) | (b&c) | (a&c);
30
31 endmodule
32
```

Constraints

```
Package x Device x adder.v x cons.xdc x
/home/mooc-03/Desktop/ADDER/ADDER.srscs/constrs_1/new/cons.xdc

1 set_property PACKAGE_PIN D9 [get_ports a]
2 set_property PACKAGE_PIN C9 [get_ports b]
3 set_property PACKAGE_PIN B9 [get_ports c]
4 set_property PACKAGE_PIN H5 [get_ports carry]
5 set_property PACKAGE_PIN J5 [get_ports sum]
6 set_property IOSTANDARD LVCMOS33 [get_ports a]
7 set_property IOSTANDARD LVCMOS33 [get_ports b]
8 set_property IOSTANDARD LVCMOS33 [get_ports c]
9 set_property IOSTANDARD LVCMOS33 [get_ports carry]
10 set_property IOSTANDARD LVCMOS33 [get_ports sum]
```

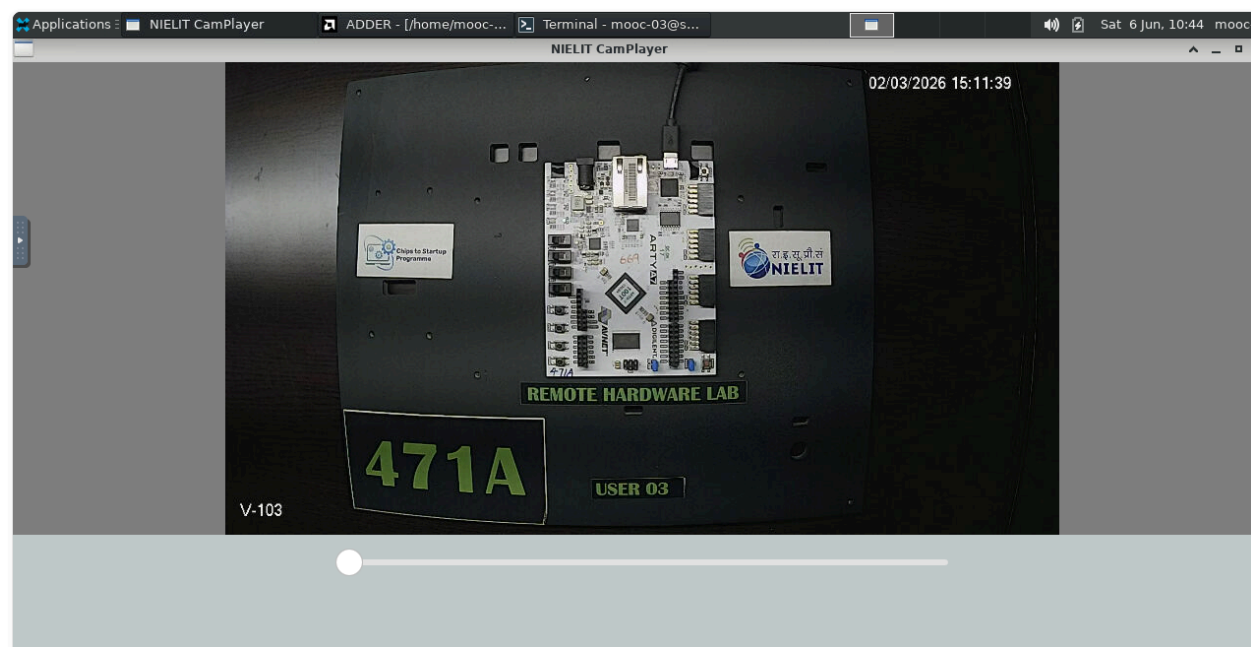
RTL

PROGRAM DEVICE

Tcl Console x Messages Serial I/O Links Serial I/O Scans ? _

```
set_property PROGRAM.FILE {/home/mooc-03/Desktop/ADDER/ADDER.runs/impl_1/adder.bit} [get_hw_devices xc7a100t_0]
current_hw_device [get_hw_devices xc7a100t_0]
refresh_hw_device -update_hw_probes false [lindex [get_hw_devices xc7a100t_0] 0]
INFO: [Labtools 27-1434] Device xc7a100t (JTAG device index = 0) is programmed with a design that has no supported debug core(s) in
set_property PROBES.FILE {} [get_hw_devices xc7a100t_0]
set_property FULL_PROBES.FILE {} [get_hw_devices xc7a100t_0]
set_property PROGRAM.FILE {/home/mooc-03/Desktop/ADDER/ADDER.runs/impl_1/adder.bit} [get_hw_devices xc7a100t_0]
program_hw_devices [get_hw_devices xc7a100t_0]
INFO: [Labtools 27-3164] End of startup status: HIGH
refresh_hw_device [lindex [get_hw_devices xc7a100t_0] 0]
INFO: [Labtools 27-1434] Device xc7a100t (JTAG device index = 0) is programmed with a design that has no supported debug core(s) in
```

CAM



Hardware ? _

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Name	Status
localhost (28)	Connected
xilinx_tcf/Digilent/210319B4A	Open
xc7a100t_0 (1)	Programmed
XADC (System Monitor)	
xilinx_tcf/Digilent/210319B4A	Closed
xilinx_tcf/Digilent/210319B4A	Closed
xilinx_tcf/Digilent/210319B4A	Closed
xilinx_tcf/Digilent/210319B4A	Closed
xilinx_tcf/Digilent/210319B4A	Closed